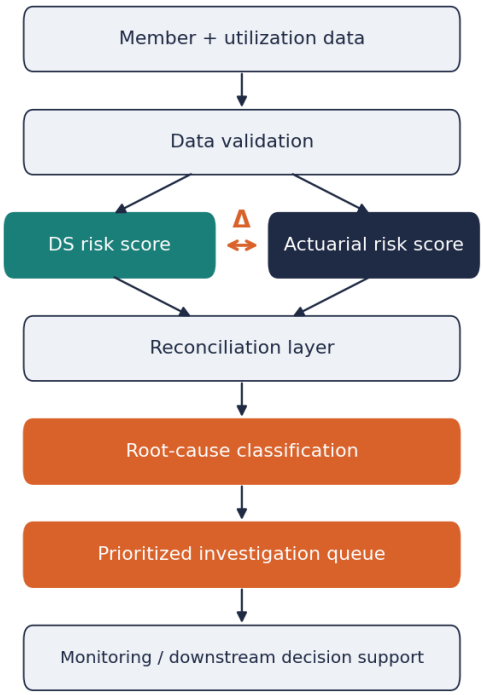
When two risk estimates disagree, how do we determine *why* — and decide what deserves investigation?

A synthetic validation and reconciliation framework, inspired by the kinds of problems described publicly by Oscar Health's Clinical Risk Adjustment Data Science team. Built with Python, SQL, pandas and statistical validation.



Don't just calculate Δ. Explain Δ.

A gap between two risk scores is a symptom, not a diagnosis. It can come from the data, the pipeline, timing, versioning, methodology — or from nothing anyone can explain yet. RiskRecon separates those causes, then ranks what to look at first.

The synthetic dataset

25,000 synthetic members — enough to show real analytical behaviour without pretending scale is the accomplishment. Two toy scoring functions score the same latent utilization.

Category	Synthetic fields
Member	member_id, age_band, region
Clinical-style	condition_count, chronic_condition_flag
Utilization	inpatient_visits, ed_visits, rx_count
Completeness	months_enrolled, claims_completeness, last_claim_date, claim-line counts
Model outputs	ds_risk_score, actuarial_risk_score
Metadata	score versions, snapshot dates

No diagnoses are modelled and no attempt is made to implement ACA HHS-HCC. The point isn't "look, I understand clinical medicine." It's **"look how I approach a consequential analytical system."**

Deliberately injected failure modes

Instead of building a dashboard, I planted six disagreement mechanisms (plus one control) so I can test whether the framework **recovers the reason I planted**. The ground truth is used only to grade the framework, never as an input to it.

#	Planted mechanism	How it is injected	Framework should say
1	**Missing-data discrepancy**	DS is built from a raw claims extract; members under 0.80 completeness lose claims	DATA_COMPLETENESS
2	**Recency discrepancy**	One side scores from an older snapshot and misses claims dated in the gap	SNAPSHOT_MISMATCH
3	**Population-segment discrepancy**	DS has diminishing returns and noise that grows with utilization	EXPECTED_MODEL_VARIANCE
4	**Transformation / version**	Staged rollout of DS v2.4, which re-defines the rx_count feature	MODEL_VERSION
5	**Pipeline / data issue**	Duplicate claim lines inflate DS utilization features	DUPLICATE_INPUT
6	**Legitimate model disagreement**	Clean data; the methodologies simply produce different estimates	EXPECTED_MODEL_VARIANCE
ctrl	**Unexplainable shocks (control)**	Idiosyncratic shocks on members with clean diagnostics	UNEXPLAINED

Scope guardrails — what this deliberately is not

Not a healthcare risk prediction model. No patient data. No claim to reproduce HHS-HCC or any proprietary model. No AUC optimized on a synthetic target. The LLM extension is optional and not the centerpiece. The impressive part should be the **reasoning and reliability architecture**, not the complexity.

Where do the models disagree?

6.9%	+0.004	0.217	0.65
of members beyond tolerance (1,727 of 25,000)	median Δ (DS − Actuarial)	mean absolute error	95th percentile |Δ|

Tolerance: a pair is in disagreement when |Δ| exceeds the larger of 0.35 points and 20% of the pair's mean score. Δ = DS score − Actuarial score.

Chart 1 — Score A vs Score B

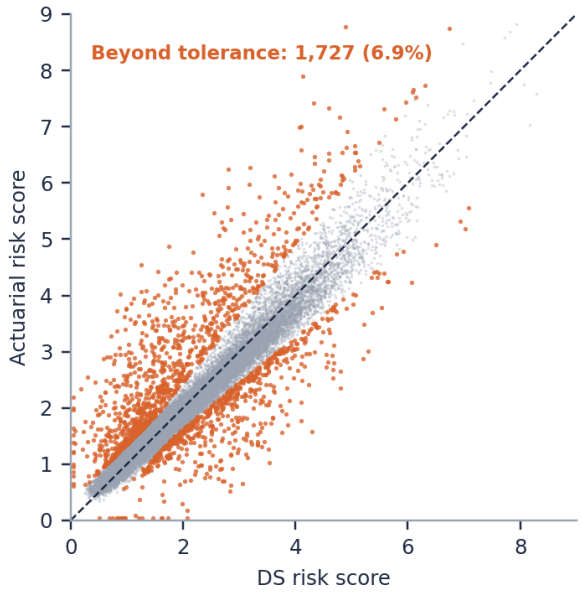


Chart 2 — Distribution of Δ

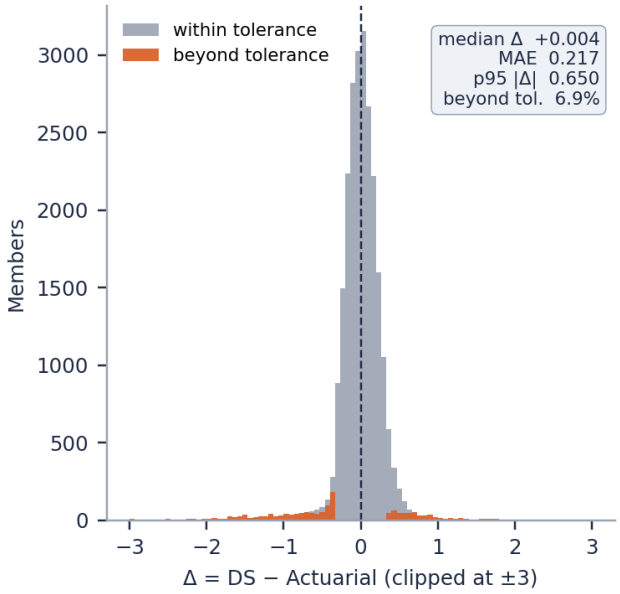
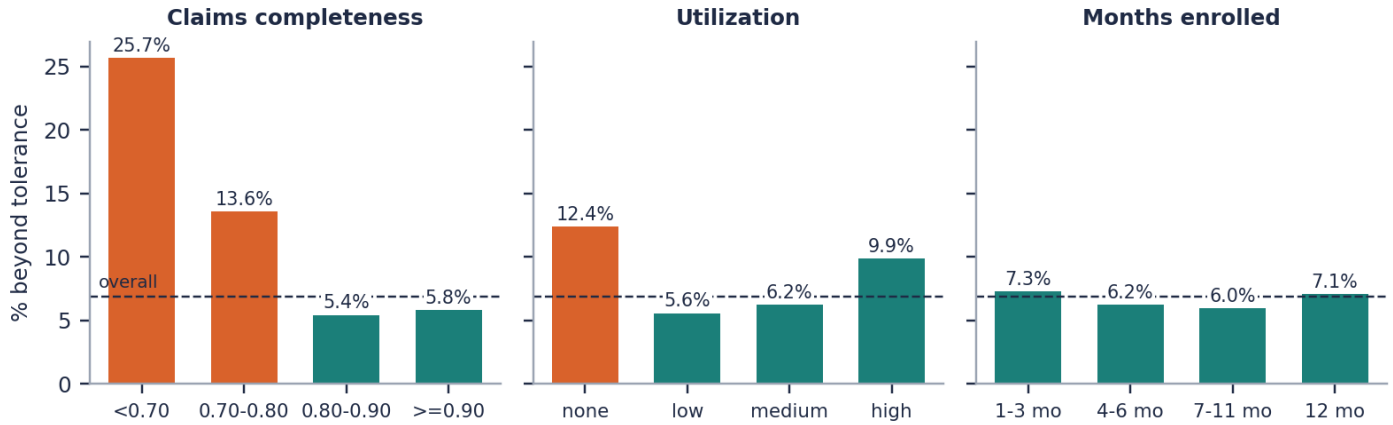


Chart 3 — Disagreement by segment: is it random, or structured?



Key finding

The largest disagreements were **not random**. Members with incomplete claims histories (completeness < 0.80) were **3.9× more likely** to exceed the reconciliation threshold (22.5% vs 5.8%; 1,660 members).

Two more things the segments say. Members with **zero observed utilization** disagree 1.8× as often as average — an empty claims history is itself a warning sign. And **enrollment duration is not a driver** (range 6.0%–7.3%): a useful negative result, because it rules out one tempting explanation.

Why do they disagree?

Every flagged member is classified. A defect is named only if its **diagnostic fires** and it **explains enough of Δ in the right direction**; otherwise the gap is compared with clean peers of similar utilization — ordinary variance, or **UNEXPLAINED**. A discrepancy isn't automatically a model failure: sometimes it's data, pipeline, timing, versioning, methodology, or genuinely unexplained.

Cause	Flagged	Mean |Δ|	Tier	Recall*	Precision*
Duplicate input DUPLICATE_INPUT	70 (4%)	0.77	**Critical**	99%	97%
Incomplete data DATA_COMPLETENESS	254 (15%)	0.73	**High**	77%	81%
Snapshot mismatch SNAPSHOT_MISMATCH	572 (33%)	1.27	**High**	100%	96%
Model version change MODEL_VERSION	243 (14%)	0.65	**High**	100%	91%
Unexplained UNEXPLAINED	232 (13%)	1.57	**Investigate**	99%	57%
Model methodology EXPECTED_MODEL_VARIANCE	356 (21%)	0.50	**Monitor**	72%	99%

*Recall/precision vs. the planted primary cause, flagged members only; overall agreement **88%**. This grades the framework on data I generated, alongside a framework I designed — it tests internal consistency, not real-world performance. Two knobs (minimum explained share 0.25; robust-z 4.0) were set after a small sensitivity grid on this data; agreement stayed between 84% and 89% across it. The misses fall into two groups: incomplete-claims cases whose unseen claims were the expensive ones (left UNEXPLAINED), and ordinary methodology gaps that coincide with a weak diagnostic and get credited to it.

Reconciliation priority score

Priority = |Δ| × Impact × Confidence (squashed to 0–1 with $1 - e^{-x}$, so the scale is fixed across snapshots)

Disagreement magnitude — how far apart are the scores? **Impact** — would the difference change a downstream use? (0.6 if the scores land in different decision bands, plus 0.4 scaled by the higher score). **Confidence** — how strongly does the evidence say something is actually wrong? (defect: 0.5 + 0.5 × share of Δ it explains; unexplained 0.6; ordinary variance 0.15).

This is my proposed framework, not a description of how any real team prioritizes work. Bands and weights are assumptions to be challenged.

The data product: an investigation queue (top 6 of 50)

#	Member	DS	Actuarial	Δ	Suspected cause	Priority
1	**M08554**	4.1	7.9	-3.8	snapshot mismatch	**0.94**
2	**M12308**	2.5	5.5	-2.9	snapshot mismatch	**0.93**
3	**M24840**	1.3	4.6	-3.3	snapshot mismatch	**0.92**
4	**M10964**	2.3	5.8	-3.5	snapshot mismatch	**0.92**
5	**M18339**	2.8	5.9	-3.1	snapshot mismatch	**0.91**
6	**M01349**	4.9	8.8	-3.9	unexplained	**0.90**

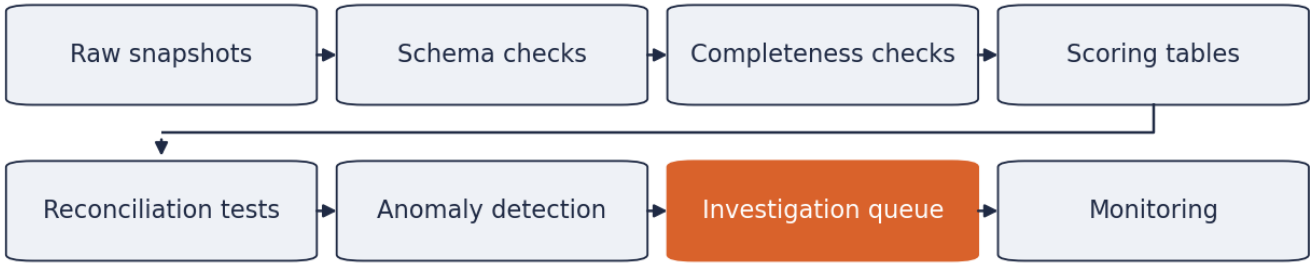
Instead of "here are 1,727 mismatched records," the output is "here are the 50 cases to investigate first, and why" (45 stale-snapshot, 5 unexplained). Stale snapshots dominate the case queue because they carry the largest, most decision-relevant gaps; **duplicate input** is tiered Critical at the *cause* level instead, because one upstream fix clears every affected member.

Highest-priority case for each root cause, with the evidence the classifier used

Member	Δ	Cause	Evidence (deterministic diagnostics)	Priority
M08554	-3.8	Snapshot mismatch	snapshot_gap_days=73 · gap_claim_load=3.86 · DS_stale	0.94
M01607	+1.0	Duplicate input	dup_claim_ratio=0.44 · delta_sign=DS_higher	0.53
M17415	-1.9	Incomplete data	claims_completeness=0.47 · delta_sign=DS_lower	0.71
M23572	+1.6	Model version change	ds_score_version=v2.4 · changed_feature=rx_count · rx_count=16	0.68
M01349	-3.9	Unexplained	robust_z=-13.0 · no_diagnostic_gate_explains_delta	0.90
M22032	-1.2	Model methodology	robust_z=-3.9 · diagnostics_clean · within clean-peer range	0.17

From analysis → dependable data product

The reconciliation is only as trustworthy as the checks around it. Validation runs before modelling, after scoring, and after classification — and a failing gate is reported, not hidden.



Data gates	
unique member ids 0 (limit == 0)	PASS
expected schema 0 (limit == 0)	PASS
acceptable missingness 0 (limit <= 0.01)	PASS
claims completeness 0.0664 (limit <= 0.05)	WARN
date freshness 0 (limit <= 0.0)	PASS
duplicate detection 0.0288 (limit <= 0.005)	FAIL
score ranges 0 (limit == 0)	PASS

Model-output gates	
score distribution drift 0.0178 (limit PSI <= 0.1)	PASS
disagreement distribution 0.65 (limit p95|delta| <= 1.0)	PASS
segment disagreement 0.257 (limit <= 0.25)	WARN
extreme value monitoring 0 (limit <= 0.001)	PASS

Reconciliation gates	
pct beyond tolerance 0.0691 (limit <= 0.1)	PASS
unexplained disagreement rate 0.0093 (limit <= 0.02)	PASS
stale snapshot rate 0.0696 (limit <= 0.02)	WARN
version mismatch rate 0.0618 (limit <= 0.02)	WARN

On this dataset: 10 pass, 4 warn, 1 fail — the failing and warning gates sit exactly where defects were planted (duplicate claim lines, incomplete claims, stale snapshots, version rollout). Without a prior snapshot, the drift gate compares the DS and actuarial score distributions to each other. The data gates are also written in SQL (sql/01_validation.sql) and tested against the Python versions.

Design principle

Knowing what a system can't explain is part of knowing that it works.

Unexplained reconciliation differences should remain visible and measurable rather than being absorbed into an aggregate performance metric or forced into a convenient label. That is why **UNEXPLAINED** is a first-class cause, why its residual is stored per member, and why it has its own gate (0.9% of members today). The framework is also graded on a planted control — shocks with clean diagnostics — that it is *supposed* to leave unexplained.

Optional extension: a Reconciliation Copilot

A small experiment, deliberately not the centerpiece. The LLM sees only structured diagnostics for one flagged case and returns a likely category, the evidence, and a next step. **It never decides whether a score is correct**; an explanation that cites evidence not in the diagnostics is rejected and replaced by the deterministic summary. The guardrail is implemented and unit-tested; **no LLM calls were made for the results in this document.**

Input (diagnostics only)	Output (must cite the input verbatim)
claims_completeness=0.47 delta=-1.90 snapshot_gap_days=0 dup_claim_ratio=0.00 ds_score_version=v2.4	category: DATA_COMPLETENESS evidence: claims_completeness=0.47 next step: re-run the DS score after claims run-out completes; compare against the completed extract.

Assumptions I'd most like to have challenged

- That DS scores are built from a raw claims extract while actuarial scores see completed claims — the completeness effect depends on it.
- The tolerance rule (larger of 0.35 points or 20% of the mean score) and the hypothetical decision bands behind Impact.
- That a defect is named once it explains about a quarter of Δ, and that a clean-peer robust-z of 4 marks "unexplained".
- Whether per-member classification is the right unit, versus tracing each cause to a single upstream fix.